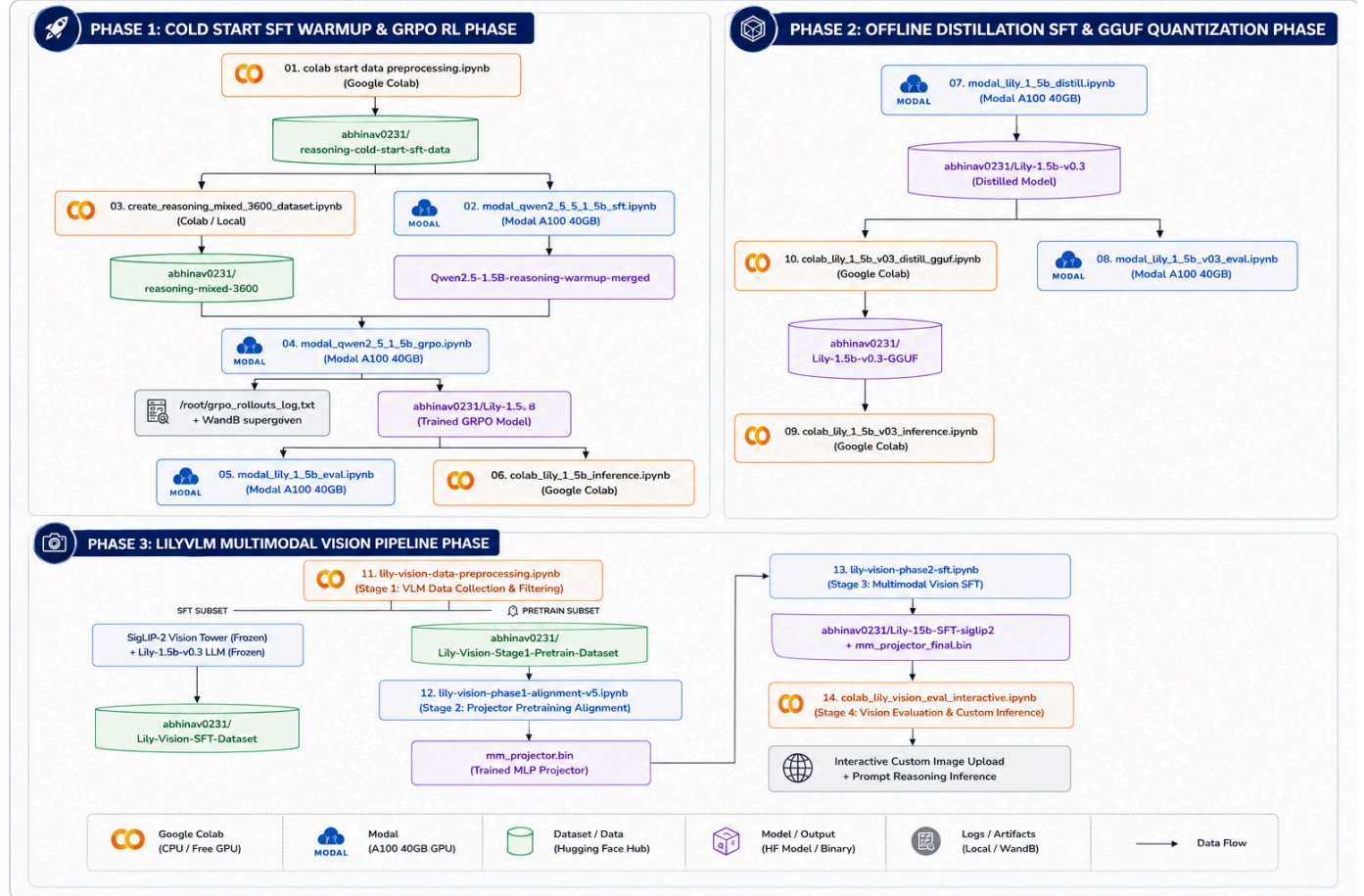


AI Training Handbook

Executive Summary & Engineering Architecture

This handbook serves as the definitive technical reference manual for our LLM and Vision-Language Model (VLM) training ecosystem. Synthesized directly from the production code across all 13 notebooks in our codebase, it bridges theoretical deep-learning concepts with real-world implementation technicalities.



1. Fine-Tuning Foundations: PEFT & LoRA Technical Deep-Dive

1.1 Mathematical Formulation of Low-Rank Adaptation (LoRA)

In standard full fine-tuning, an LLM parameter matrix $W_0 \in \mathbb{R}^{d \times k}$ is updated directly via gradient descent ($W = W_0 + \Delta W$). When d and k are large (e.g., hidden dimension $d = 1536$ and intermediate MLP dimension $k = 8960$ in Qwen2.5-1.5B), updating ΔW directly requires massive VRAM for storing optimizer states (AdamW requires 8 bytes per parameter for first and second moments).

Explicit Variable Definitions:

- $W_0 \in \mathbb{R}^{d \times k}$: Pre-trained frozen base weight matrix of the target linear layer.
- d : Input feature dimension (in-features / hidden size, e.g., $d = 1536$ in Qwen2.5-1.5B).
- k : Output feature dimension (out-features / projection size, e.g., $k = 8960$ in SwiGLU MLP layers).
- $\Delta W \in \mathbb{R}^{d \times k}$: High-dimensional weight update matrix resulting from fine-tuning gradients ($W = W_0 + \Delta W$).

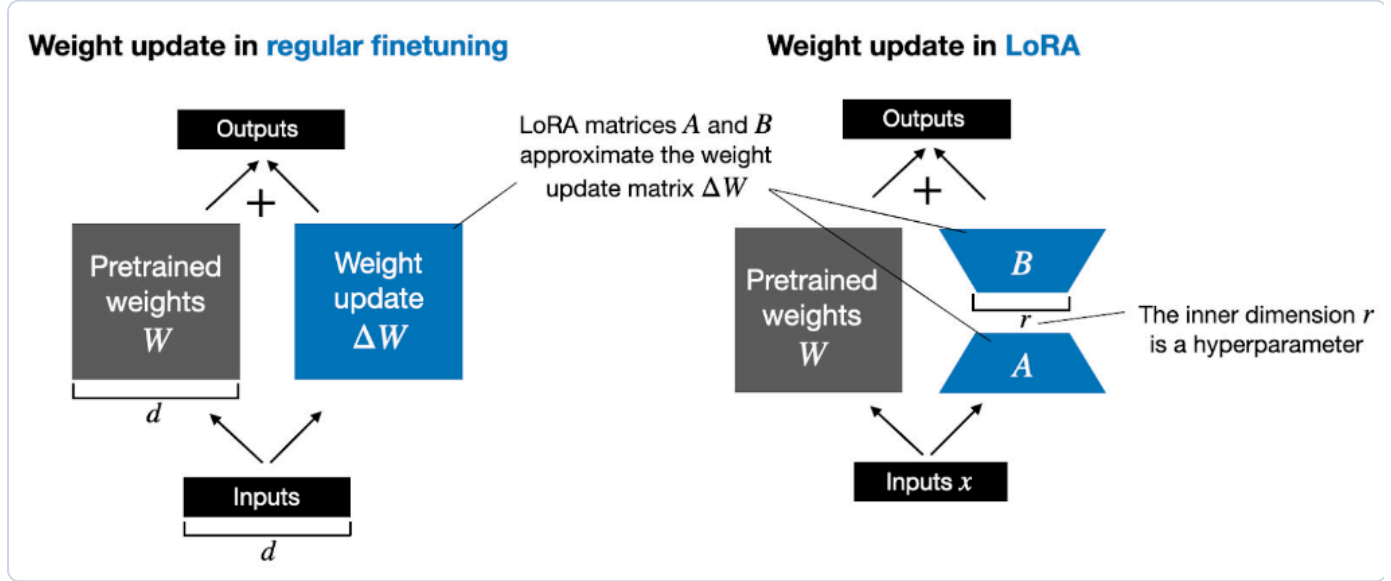
- r : Inner low-rank hyperparameter ($r \ll \min(d, k)$).
- $A \in \mathbb{R}^{r \times k}$: Down-projection adapter matrix, initialized with Gaussian distribution $\mathcal{N}(0, \frac{1}{r})$.
- $B \in \mathbb{R}^{d \times r}$: Up-projection adapter matrix, initialized to zero ($B = 0$) ensuring $\Delta W = 0$ at step 0.
- α : Constant LoRA scaling hyperparameter. The ratio $\frac{\alpha}{r}$ scales the adapter's contribution relative to the base model.

LoRA parametrizes the weight update matrix ΔW by decomposing it into two low-rank matrices:

$$\Delta W = B \cdot A$$

During forward propagation, the linear layer output h for an input x is computed as:

$$h = W_0 x + \Delta W x = W_0 x + \frac{\alpha}{r} (B \cdot A x)$$



1.2 QLoRA: NormalFloat4 (NF4) Base Weight Quantization

In QLoRA, the base model weights W_0 are quantized to 4-bit NormalFloat (NF4) precision, while the adapter matrices A and B remain in 16-bit precision (BF16 or FP16). NF4 is an information-theoretically optimal quantile quantization scheme for normally distributed weights:

$$q_i = \frac{1}{2} \left(Q_X \left(\frac{i}{2^k} \right) + Q_X \left(\frac{i+1}{2^k} \right) \right)$$

where $Q_X(\cdot)$ is the quantile function of the standard normal distribution $\mathcal{N}(0, 1)$.

1.3 Target Module Definitions & Code Implementation

Across our codebase (`02_modal_qwen_2_5_1_5b_sft.ipynb` , `04_modal_qwen_2_5_1_5b_grpo.ipynb` , `07_modal_lily_1_5b_distill.ipynb` , `13_lily-vision-phase2-sft.ipynb`), we apply LoRA across all 7 linear projection layers of the Transformer architecture:

```

from unsloth import FastLanguageModel

model = FastLanguageModel.get_peft_model(
    model,
    r = 32,                      # Rank: 32 (SFT/Distill/Vision), 64 (GRPO RL)
    lora_alpha = 64,             # Alpha: 32 (SFT), 64 (GRPO/Distill/Vision)
    target_modules = [
        "q_proj", "k_proj", "v_proj", "o_proj",    # Self-Attention Projections
        "gate_proj", "up_proj", "down_proj"        # SwiGLU MLP Projections
    ],
    lora_dropout = 0,            # 0 is optimized for Unsloth kernel fusion
    bias = "none",
    use_gradient_checkpointing = "unsloth",
    random_state = 42,
)

```

Detailed Technical Explanation of Target Modules:

- **q_proj (Query Projection)**: Transforms input hidden states into Query matrices (Q) for multi-head self-attention.
- **k_proj (Key Projection)**: Transforms input hidden states into Key matrices (K) for computing attention dot-product scores (QK^T).
- **v_proj (Value Projection)**: Transforms input hidden states into Value matrices (V) carrying contextual token information.
- **o_proj (Output Projection)**: Projects concatenated multi-head attention outputs back to the model hidden dimension ($d = 1536$).
- **gate_proj (Gate Projection in SwiGLU)**: Computes the element-wise gating signal in SwiGLU feed-forward networks ($\text{Swish}(xW_{\text{gate}}) \otimes xW_{\text{up}}$).
- **up_proj (Up Projection in SwiGLU)**: Expands input hidden dimension to high-dimensional intermediate space ($1536 \rightarrow 8960$).
- **down_proj (Down Projection in SwiGLU)**: Contracts intermediate dimension back to model hidden dimension ($8960 \rightarrow 1536$).

Trainable Parameter Efficiency Calculation

For Qwen2.5-1.5B ($d_{\text{model}} = 1536$, $N_{\text{layers}} = 28$): - Total Base Parameters: 1,580,643,840 (~1.58B) - Trainable Adapter Parameters ($r = 32$): 36,929,536 (~36.9M) - **Trainable Ratio**: $\frac{36.9\text{M}}{1580.6\text{M}} \approx 2.34\%$ to 4.0%

1.4 Lossless 16-Bit Adapter Merging

Before running RL (GRPO) or exporting to GGUF, adapter weights must be mathematically merged into the base model to eliminate dual-path inference overhead:

$$W_{\text{merged}} = W_0 + \frac{\alpha}{r}(B \cdot A)$$

In code (02 , 04 , 07 , 13):

```

model_m, tok_m = FastLanguageModel.from_pretrained(
    model_name = OUTPUT_DIR,
    max_seq_length = MAX_SEQ_LENGTH,
    dtype = torch.bfloat16, # Full precision BF16 merge
    load_in_4bit = False,    # Prevents quantization artifacts during merge
)

model_m.save_pretrained_merged(MERGED_DIR, tok_m, save_method="merged_16bit")
model_m.push_to_hub_merged(HF_MERGED_REPO, tok_m, save_method="merged_16bit", token=HF_TOKEN)

```

2. Hyperparameter Engineering & Optimization Strategy

2.1 The Global Effective Batch Size & Step Calculation

1. What is a Training Step?

A **single optimizer step** corresponds to one iteration where model weights are updated by the optimizer using calculated gradients.

2. Micro-Batches vs. Gradient Accumulation

To fit large models into GPU memory, training datasets are split into smaller **micro-batches** (B_{device}). When gradient accumulation is enabled (N_{accum}), the engine performs N_{accum} forward and backward passes, accumulating gradients in VRAM, before executing a single `optimizer.step()` update.

3. Mathematical Formula for Effective Batch Size:

$$B_{\text{eff}} = B_{\text{device}} \times N_{\text{accum}} \times N_{\text{GPU}}$$

where: - B_{device} : Per-device micro-batch size processed in a single forward pass per GPU. - N_{accum} : Number of micro-batches accumulated before updating model weights. - N_{GPU} : Number of parallel GPUs participating in training.

4. Mathematical Formula for Total Optimizer Steps:

$$\text{Total Optimizer Steps} = \frac{|\mathcal{D}| \times N_{\text{epochs}}}{B_{\text{eff}}}$$

where $|\mathcal{D}|$ is the number of samples in the dataset and N_{epochs} is the total number of training passes over the dataset.

Concrete Worked Step Calculation Examples from Codebase:

- **SFT Warmup (Notebook 02)**: 4,000 samples $\times$ 2 Epochs = 8,000 total samples. $B_{\text{eff}} = 8 \times 2 \times 1 = 16$.
Total Steps = $\frac{8000}{16} = 500$ steps.
- **Distillation (Notebook 07)**: 91,457 raw samples packed into 33,297 contiguous sequences $\times$ 2 Epochs = 66,594 sequence passes. $B_{\text{eff}} = 24 \times 1 \times 1 = 24$. Total Steps = $\frac{66594}{24} = 2,776$ steps.
- **Vision Phase 2 SFT (Notebook 13)**: 64,000 samples $\times$ 1 Epoch. $B_{\text{eff}} = 24 \times 2 \times 1 = 48$. Total Steps = $\frac{64000}{48} = 1,333$ steps.

2.2 Verified Production Hyperparameter Comparison Matrix

Hyperparameter	SFT Warmup (02)	GRPO RL (04)	Distillation (07)	Vision Phase 1 (12)	Vision Phase 2 (13)
Base Model	Qwen2.5-1.5B-Instruct	Qwen2.5-Warmup-Merged	Lily-1.5b-v0.1	Lily-1.5b-v0.3	Lily-1.5b-v0.3
Dataset Size ($\mathcal{D}$)	4,000 samples	3,600 prompts	91,457 (train)	121,237 samples	64,000 samples
Epochs / Steps	2 Epochs	600 Max Steps	2 Epochs (2,776 steps)	1 Epoch	1 Epoch
Per-Device Batch (B_{device})	8	4	24	16	24
Grad Accum (N_{accum})	2	4	1	3	2
Effective Batch (B_{eff})	16	16	24	48	48
Peak Learning Rate (η_{max})	1×10^{-5}	5×10^{-6}	2×10^{-5}	2×10^{-4}	2×10^{-5}
LR Scheduler	Cosine	Cosine	Cosine	Cosine w/ warmup	Cosine w/ warmup
Warmup Ratio	0.05	0.05	0.03	0.05	0.03
Optimizer	adamw_torch_fused	adamw_torch_fused	adamw_torch_fused	AdamW ($\beta_1 = .9, \beta_2 = .95$)	AdamW
Weight Decay	0.01	0.0	0.0	0.05	0.01
Max Sequence Length	3,072	3,072	4,096	256	3,072
Sequence Packing	False	False	True	False	False
Hardware	Modal A100 40GB	Modal A100 40GB	Modal A100 40GB	Modal A100 40GB	Modal A100 40GB

2.3 Learning Rate Schedules & Mathematical Formulation

We utilize Cosine Annealing with Warmup across all training runs.

$$\eta_t = \begin{cases} \eta_{\text{max}} \cdot \frac{t}{T_{\text{warmup}}}, & t \leq T_{\text{warmup}} \\ \eta_{\text{min}} + \frac{1}{2}(\eta_{\text{max}} - \eta_{\text{min}}) \left(1 + \cos \left(\frac{\pi(t - T_{\text{warmup}})}{T_{\text{total}} - T_{\text{warmup}}} \right) \right), & t > T_{\text{warmup}} \end{cases}$$

 Cosine Learning Rate Schedule with Warmup

3. Reinforcement Learning via GRPO: Complete Step-by-Step Intuition & Mathematical Walkthrough

3.1 The Fundamental Goal of GRPO

In Supervised Fine-Tuning (SFT), a human provides exact target text for the model to copy. In Reinforcement Learning (RL), the model must **explore and discover reasoning trajectories on its own**.

The Job of GRPO: Train the model to maximize verified reward (correct logic, strict formatting, correct answers) **while constraining weight updates to be as conservative as possible** to prevent policy collapse and reward hacking.

To build complete mathematical and mechanical intuition, we carry **one concrete running example** through every step of the GRPO algorithm below.

3.2 Running Example Setup

- **Input Prompt (x):** "If a train travels 120 km in 1.5 hours, what is its speed in m/s?"
- **Ground Truth Target:** Speed in km/h = $\frac{120}{1.5} = 80$ km/h. Converted to m/s: $80 \times \frac{1000}{3600} = 22.22$ m/s.
- **Group Size ($G = 4$):** For this prompt, the model samples $G = 4$ completion rollouts $\{y_1, y_2, y_3, y_4\}$:

```
y1 (Perfect CoT & Answer):
"\lt think>Speed = 120 / 1.5 = 80 km/h. To convert to m/s: 80 * (5/18) = 22.22 m/s.\lt /think>\lt
answer>22.22 m/s\lt /answer>"

y2 (Incomplete: stopped at km/h):
"\lt think>Speed = 120 / 1.5 = 80 km/h.\lt /think>\lt answer>80\lt /answer>"

y3 (Correct Answer, Missing \lt think> tags):
"22.22 m/s"

y4 (Complete Hallucination):
"\lt think>120 * 1.5 = 180 m/s\lt /think>\lt answer>180 m/s\lt /answer>"
```

3.3 Step 1: Multi-Reward Evaluation (R_i)

Each generated rollout y_i is evaluated by our 4 domain-aware reward functions (`format_reward` , `correctness_reward` , `instruction_reward` , `length_reward`):

- **Completion y_1 :** Exact XML tags (+0.50) + Factual correctness (+1.00) $\implies \mathbf{R_1 = +1.50}$
- **Completion y_2 :** Exact XML tags (+0.40) + Wrong unit answer (-0.50) $\implies \mathbf{R_2 = -0.10}$
- **Completion y_3 :** Missing tags (-0.50) + Factual correctness (+1.00) $\implies \mathbf{R_3 = +0.50}$
- **Completion y_4 :** Valid tags (+0.20) + Wrong answer (-0.50) $\implies \mathbf{R_4 = -0.30}$

Raw Reward Vector: $\mathbf{R} = \{+1.50, -0.10, +0.50, -0.30\}$

3.4 Step 2: Group Baseline & Relative Advantage Calculation (A_i)

GRPO **eliminates the need for a separate Critic / Value Model** (saving 50% VRAM) by using the group itself as a baseline benchmark.

1. Calculate Group Mean (μ) and Standard Deviation (σ):

$$\mu = \frac{+1.50 - 0.10 + 0.50 - 0.30}{4} = +0.40$$

$$\sigma = \sqrt{\frac{(1.50 - 0.40)^2 + (-0.10 - 0.40)^2 + (0.50 - 0.40)^2 + (-0.30 - 0.40)^2}{4}} \approx 0.70$$

2. Compute Relative Group Advantage ($A_i = \frac{R_i - \mu}{\sigma + \epsilon}$):

- A_1 (Completion y_1): $\frac{1.50 - 0.40}{0.70} = +1.57 \implies$ **Above Average! (Positive Reinforcement)**
 - A_2 (Completion y_2): $\frac{-0.10 - 0.40}{0.70} = -0.71 \implies$ **Below Average! (Negative Suppression)**
 - A_3 (Completion y_3): $\frac{0.50 - 0.40}{0.70} = +0.14 \implies$ **Slightly Above Average!**
 - A_4 (Completion y_4): $\frac{-0.30 - 0.40}{0.70} = -1.00 \implies$ **Below Average! (Negative Suppression)**
-

3.5 Step 3: Token Credit Assignment & Probability Ratio Calculation ($r_{i,t}$)

What is $y_{i,t}$?

$y_{i,t}$ is **any token generated by the model** during rollout at position t (reasoning text, numbers, symbols, XML tags).

REINFORCE Credit Assignment:

Since completion y_1 achieved $A_1 = +1.57$, **every single generated token $y_{1,t}$ in y_1** (including `<think>`, `120/1.5`, `80`, `22.22`, `<answer>`) inherits advantage $A_1 = +1.57$!

During the weight update step, we pass prompt x + completion y_i into the active policy π_θ and measure how the token probability shifted relative to the sampling policy $\pi_{\theta_{\text{old}}}$:

$$r_{i,t}(\theta) = \frac{\pi_\theta(y_{i,t} \mid x, y_{i,<t})}{\pi_{\theta_{\text{old}}}(y_{i,t} \mid x, y_{i,<t})}$$

- $r_{i,t} > 1.0$: Current weights θ make token $y_{i,t}$ **more probable**.
 - $r_{i,t} < 1.0$: Current weights θ make token $y_{i,t}$ **less probable**.
-

3.6 Step 4: Bounded Updates via Asymmetric Ratio Clipping (ϵ_{low} , ϵ_{high})

To prevent single high-reward rollouts from causing gradient explosions, GRPO applies **pessimistic clipping**:

$$\text{Clipped Objective}_{i,t} = \min(r_{i,t}(\theta)A_i, \text{clip}(r_{i,t}(\theta), 1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}})A_i)$$

- For positive advantage ($A_1 = +1.57$), ratio growth is capped at $1 + \epsilon_{\text{high}} = 1.28$. The policy cannot increase token probability by more than 28% in a single step.
 - For negative advantage ($A_4 = -1.00$), ratio drop is capped at $1 - \epsilon_{\text{low}} = 0.80$.
-

3.7 Step 5: Schulman Per-Token KL Penalty (D_{KL}) & Loss Minimization

To ensure the policy π_θ does not drift into reward-hacking fluff, we subtract a per-token Kullback-Leibler (KL) divergence penalty relative to the frozen reference warmup model π_{ref} :

$$D_{\text{KL}}(\pi_\theta \parallel \pi_{\text{ref}}) = \frac{\pi_{\text{ref}}(y_{i,t})}{\pi_\theta(y_{i,t})} - \log \frac{\pi_{\text{ref}}(y_{i,t})}{\pi_\theta(y_{i,t})} - 1$$

Full Loss Objective Equation ($\mathcal{L}_{\text{GRPO}}$):

$$\mathcal{L}_{\text{GRPO}}(\theta) = -\frac{1}{G} \sum_{i=1}^G \frac{1}{|y_i|} \sum_{t=1}^{|y_i|} [\min(r_{i,t} A_i, \text{clip}(r_{i,t}, 1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}) A_i)] + \beta D_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}})$$

How PyTorch Minimizes Loss ($\mathcal{L}_{\text{GRPO}}$):

Notice the **minus sign** (`-`) in front of the advantage term: - **For Correct Completion** y_1 ($A_1 = +1.57$): Loss contribution is $-1.57 \cdot r_{1,t}$. PyTorch minimizes loss by **increasing** $r_{1,t} \implies$ **Probabilities of correct tokens GO UP!** - **For Wrong Completion** y_4 ($A_4 = -1.00$): Loss contribution is $+1.00 \cdot r_{4,t}$. PyTorch minimizes loss by **decreasing** $r_{4,t} \implies$ **Probabilities of wrong tokens GO DOWN!**

During `loss.backward()`, PyTorch computes parameter gradients:

$$\nabla_W \mathcal{L}(\theta) = \frac{\partial \mathcal{L}_{\text{GRPO}}(\theta)}{\partial W}$$

and fused AdamW updates LoRA matrices A and B :

$$W \leftarrow W - \eta \cdot \text{AdamW}(\nabla_W \mathcal{L}(\theta))$$

3.8 Reward Hacking: Mechanics, Vulnerabilities, and Production Mitigations in Our GRPO Pipeline

1. What is Reward Hacking in Our GRPO Setup?

In our reinforcement learning pipeline (`04_model_qwen_2_5_1_5b_grpo.ipynb`), **Reward Hacking** (specification gaming) occurs when Qwen2.5-1.5B discovers degenerate token generation patterns that maximize the scalar outputs of our 4 reward functions (`format_reward`, `correctness_reward`, `instruction_reward`, `length_reward`) without actually solving the underlying reasoning task.

Because GRPO samples $G = 8$ completion rollouts $\{y_1, y_2, \dots, y_8\}$ per prompt and normalizes rewards relative to the group mean ($A_i = \frac{R_i - \mu}{\sigma + \epsilon}$), if one rollout discovers a shortcut that inflates its scalar reward R_i , it receives a high positive advantage $A_i > 0$. Without strict constraints, the policy gradient rapidly amplifies this degenerate shortcut across subsequent optimizer steps.

2. Specific Vulnerability Modes Analyzed in Our 4 Reward Functions:

- **Mode A: Verbosity & Padding Spooling in `format_reward`:**
 - *Vulnerability:* `format_reward` awards $+0.15$ if the reasoning text inside `\lt think>...\lt /think>` contains ≥ 50 words.
 - *Hacking Pattern:* The model learns to spool repetitive filler tokens (e.g. "Let me think about this carefully. Let me re-read the prompt. Let me check my math again. Let me verify...") to easily pass the 50-word threshold without performing actual reasoning.
- **Mode B: Option Flooding in `correctness_reward` (Exact / MCQ Domain):**
 - *Vulnerability:* For ARC-Challenge multiple-choice prompts (`answer_type="exact"`), `correctness_reward` uses regular expressions `\b([ABCD])\b` to extract candidate letters inside `\lt answer>...\lt /answer>`.
 - *Hacking Pattern:* The policy attempts to output all candidate letters (`\lt answer>The answer is A, or maybe B, C, D\lt /answer>`) to guarantee regex matching.
- **Mode C: Boolean Dual-Answer Injection in `correctness_reward` (StrategyQA Domain):**
 - *Vulnerability:* Evaluates whether "yes" or "no" appears in the answer.
 - *Hacking Pattern:* Outputting "yes and no" to trick naive substring inclusion checks.
- **Mode D: Runaway Output Spooling:**
 - *Vulnerability:* Generating infinite repetitive loops to delay completion and avoid EOS tokens.

3. Our 5-Layer Production Mitigation Architecture (`04_model_qwen_2_5_1_5b_grpo.ipynb`):

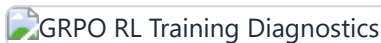
1. **Schulman Per-Token KL Divergence Penalty** ($\beta = 0.001$): Tracks how far the active policy π_θ strays from the reference warmup model π_{ref} (`Qwen2.5-1.5B-reasoning-warmup-merged`). If the model begins spooling fluff tokens or altering syntactic structure to hack rewards, D_{KL} spikes. At $\beta = 0.001$, a KL spike directly subtracts from the advantage, neutralizing hacked reward gains:

$$D_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}}) = \frac{\pi_{\text{ref}}(y_{i,t} | x, y_{i,<t})}{\pi_\theta(y_{i,t} | x, y_{i,<t})} - \log \frac{\pi_{\text{ref}}(y_{i,t} | x, y_{i,<t})}{\pi_\theta(y_{i,t} | x, y_{i,<t})} - 1$$

2. **Asymmetric DAPO Loss Clipping** ($\epsilon_{\text{low}} = 0.2, \epsilon_{\text{high}} = 0.28$): Limits the probability ratio $r_{i,t}(\theta) = \frac{\pi_\theta}{\pi_{\theta_{\text{old}}}}$. Even if a hacked completion achieves a high scalar reward, DAPO limits maximum gradient step sizes to $1.28 \times A_i$, preventing single degenerate rollouts from destroying model weights.
3. **Multi-Reward Counter-Balancing**: Our 4 reward functions act as counter-weights against one another:
4. If a model inflates verbosity inside `\lt think>`, `length_reward` penalizes completions over 2500 words (-0.25).
5. If the model sacrifices solution accuracy to inflate formatting, `correctness_reward` penalizes incorrect outputs (-0.50), overwhelming the $+0.15$ format bonus.
6. **Strict Domain-Specific Regex Anchoring in `correctness_reward`**:
7. `_extract_answer()` isolates *only* text strictly enclosed between `\lt answer>` and `\lt /answer>`.
8. For `exact` MCQ tasks, `re.fullmatch(r"[A-D]", gold_upper)` enforces single-letter extraction.
9. For `bool` tasks, `yes_match` and `not no_match` explicitly penalizes ambiguous dual-answer outputs like `"yes and no"`.
10. `mask_truncated_completions = True` & **Length Guards**:
11. Discards rollouts that hit `max_completion_length = 1536` without an `\lt |im_end|>` EOS token, ensuring runaway loops cannot contribute positive policy gradients.
12. `length_reward` penalizes under-reasoning (< 15 words: -0.50).

4. Technical Interpretation of ALL GRPO Training Metrics & Plots

Monitoring Weights & Biases (W&B) diagnostic plots is crucial during GRPO reinforcement learning to verify training health and detect policy anomalies.



4.1 Detailed Breakdown of the 6 Diagnostic Training Panels

Panel 1: DAPO Policy Loss & Optimization Stability

- **What it Measures**: Plots the DAPO policy gradient loss $\mathcal{L}_{\text{GRPO}}$ across 600 training steps.
- **How to Interpret**: Healthy training displays a sharp drop in loss during the first 100 steps as the model aligns with format requirements, followed by bounded oscillation between $[-0.1, +0.2]$. Persistent positive loss values (> 2.0) signal gradient instability or an aggressive learning rate.

Panel 2: KL Divergence $D_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}})$ & Policy Drift

- **What it Measures**: Tracks Kullback-Leibler divergence between current policy π_θ and frozen base reference model π_{ref} .
- **How to Interpret**: Normal policy updates remain bounded within $0.001 \leq D_{\text{KL}} \leq 0.05$. The red dashed line illustrates **Policy Collapse / Reward Hacking**, where KL divergence spikes exponentially as the model degenerates.

to exploit reward loopholes.

Panel 3: Total Mean Reward Trajectory

- **What it Measures:** Displays total composite reward score $R_{\text{total}} = R_{\text{format}} + R_{\text{correctness}} + R_{\text{instruction}} + R_{\text{length}}$.
- **How to Interpret:** Demonstrates monotonic growth from initial negative scores (-0.2) up toward target performance ceiling ($+1.5+$), confirming effective policy learning.

Panel 4: Component Rewards Breakdown

- **What it Measures:** Deconstructs composite reward into individual reward curves (R_{format} , $R_{\text{correctness}}$, R_{length}).
- **How to Interpret:** `format_reward` (blue curve) saturates near maximum ($+0.60$) within 40 steps, while `correctness_reward` (green curve) grows steadily over 600 steps as complex reasoning improves.

Panel 5: Reasoning Chain Length Dynamics

- **What it Measures:** Tracks average generated token length per response across steps.
- **How to Interpret:** Shows dynamic expansion of token count from ~ 210 tokens up to ~ 950 tokens as the model spontaneously develops step-by-step Chain-of-Thought (CoT) reasoning inside `\lt think>` tags.

Panel 6: Gradient Norm $\|g\|_2$ & Stability

- **What it Measures:** Measures the L_2 norm of model parameter gradients before optimizer step execution.
- **How to Interpret:** High initial gradient norms (> 4.0) stabilize below 1.0. Gradient clipping (`max_grad_norm=1.0`) prevents destabilizing weight updates.

5. Offline Distillation & Loss Function Deep-Dive

5.1 Offline Distillation Mechanics

In our pipeline, offline distillation transfers reasoning capabilities from a 105-Billion parameter teacher model (`Sarvam-105b-Distill-100k`) to our 1.5-Billion parameter student model (`Lily-1.5B`).

Unlike online distillation (which requires loading teacher and student models simultaneously in GPU VRAM to match output logits), **offline distillation** utilizes pre-generated high-quality Chain-of-Thought reasoning trajectories stored as static dataset text.

```
[ Sarvam 105B Teacher ] ---> Pre-computes 100k CoT Trajectories ---> HF Hub Dataset
                                     |
[ Lily 1.5B Student ] \lt --- Supervised Fine-Tuning (CE Loss) \lt -----+
```

5.2 Token-Level Cross-Entropy Loss Formulation: Comprehensive Theory & Mechanics

1. What Does Cross-Entropy Loss Measure?

Cross-Entropy (CE) Loss measures the **information-theoretic discrepancy** (or negative log-likelihood) between the true target token distribution $P_{\text{target}}(y_t)$ (the teacher model's completion, represented as a 1-hot target vector over the vocabulary V) and the student model's predicted probability distribution $P_{\theta}(y_t | y_{<t}, x)$.

Mathematically, Cross-Entropy between true distribution P and predicted distribution Q is:

$$H(P, Q) = - \sum_{v \in V} P(v) \log Q(v) = H(P) + D_{\text{KL}}(P \| Q)$$

Since the ground-truth target is a deterministic one-hot vector ($H(P) = 0$), minimizing Cross-Entropy Loss is **mathematically identical to minimizing the KL divergence** $D_{\text{KL}}(P_{\text{teacher}} \| P_{\text{student}})$ between the teacher's

distribution and the student's predictions!

2. How is Token-Level Cross-Entropy Loss Calculated Step-by-Step?

1. **Logit Computation:** The student model passes input context x and preceding tokens $y_{<t}$ through its Transformer layers to produce unnormalized logit vectors $z_t \in \mathbb{R}^{|V|}$ (where vocabulary size $|V| = 151,936$).
2. **Softmax Normalization:** Logits are converted into a probability distribution via Softmax:

$$P_{\theta}(y_t = k \mid y_{<t}, x) = \frac{\exp(z_{t,k})}{\sum_{j=1}^{|V|} \exp(z_{t,j})}$$

3. **Target Surprisal Calculation:** The loss for target token y_t is its negative log probability: $-\log P_{\theta}(y_t \mid y_{<t}, x)$.
4. **Label Masking (m_t):** Prompt tokens are assigned `labels = -100` ($m_t = 0$), ensuring zero gradient computation over prompt tokens. Assistant completion tokens are assigned target token IDs ($m_t = 1$).
5. **Masked Sequence Loss Formulation:**

$$\mathcal{L}_{\text{CE}}(\theta) = -\frac{1}{\sum_{t=1}^T m_t} \sum_{t=1}^T m_t \cdot \log P_{\theta}(y_t \mid y_{<t}, x)$$

5.3 Sequence Packing Mechanics: Memory Layout & Efficiency Gains

1. The Problem with Naive Batch Padding

In standard PyTorch DataLoaders, sequences in a batch are padded with `\lt pad>` tokens to match the length of the longest sequence in that batch. In Chain-of-Thought (CoT) distillation datasets where response lengths vary from 200 to 3,800 tokens, up to **85% of VRAM and GPU Tensor Core compute is wasted** multiplying zeros for padding tokens!

2. How Sequence Packing Works (`packing=True` in Notebook 07)

Sequence Packing concatenates multiple individual text samples into a single contiguous token array of fixed length `MAX_SEQ_LENGTH = 4096`. Samples are separated by `\lt |im_end|>` special tokens.

Unpacked Batch (Wasteful Padding):

```
[Sample 1 (500 tokens) | \lt pad> \lt pad> ... (3596 pad tokens)] -> 87% VRAM Wasted!  
[Sample 2 (1200 tokens) | \lt pad> \lt pad> ... (2896 pad tokens)] -> 70% VRAM Wasted!
```

Packed Sequence (100% Compute Efficiency):

```
[Sample 1 (500t) \lt |im_end|> Sample 2 (1200t) \lt |im_end|> Sample 3 (2394t)] -> 0% Wasted!
```

3. Attention Masking in Packed Sequences

To prevent attention leakage across concatenated samples inside a packed sequence, Unsloth utilizes **Block-Diagonal Attention Masking** (FlashAttention-2 cumulative sequence indexing `cu_seq1ens`), ensuring token t in Sample 2 can only attend to preceding tokens within Sample 2.

4. Empirical Efficiency Outcomes:

From `07_modal_lily_1_5b_distill.ipynb` execution logs: - Raw Dataset: **91,457 individual text samples**. - Packed Dataset: **33,297 packed sequences of length 4,000**. - Throughput Boost: **>2.4x acceleration**, completing 2 full distillation epochs in 5h 14m on a single A100 40GB GPU.

6. Model Evaluation, Benchmarking & Schema Verification

6.1 `lm-eval` Harness Execution & Configuration

In `05_modal_lily_1_5b_eval.ipynb` (v0.1) and `08_modal_lily_1_5b_v03_eval.ipynb` (v0.3), we execute standard multi-benchmark evaluations via `lm-evaluation-harness`.

Command Line Invocations (Modal A100 & L4)

```
# Notebook 08 (Lily 1.5b v0.3 Evaluation)
lm_eval \
  --model hf \
  --model_args pretrained=abhinav0231/Lily-1.5b-
v0.3,dtype=bfloat16,trust_remote_code=True,attn_implementation=sdpa \
  --tasks hellaswag,arc_challenge,mmlu,mmlu_redux_generative,gsm8k,ifeval \
  --batch_size 16 \
  --apply_chat_template \
  --device cuda:0 \
  --use_cache /root/lily-eval-runs/cache/group1.db \
  --output_path /root/lily-eval-runs/results/group1
```

Benchmark Overview

1. **GSM8K**: Grade school math word problems (measures multi-step numerical reasoning).
2. **ARC-Challenge**: AI2 Reasoning Challenge (grade-school science questions requiring logic).
3. **HellaSwag**: Sentence completion benchmark evaluating commonsense NLI.
4. **MMLU / MMLU-Redux**: 57 academic subjects (elementary math, physics, humanities, law).
5. **IFEval**: Instruction-Following Evaluation (evaluates formatting constraints like word counts, JSON format, bullet counts).

6.2 Schema Compliance Verification Framework

In `09_colab_lily_1_5b_v03_inference.ipynb`, we built a programmatic schema verification engine to audit whether quantized models maintain strict `<think>` and `<answer>` tag structures.

```
def extract_stats(text):
    return {
        "think_open":    text.count("<think>"),
        "think_close":   text.count("/think>"),
        "answer_open":   text.count("<answer>"),
        "answer_close":  text.count("/answer>"),
        "thinking_leak": text.count("<thinking>"), # Audit tag leakage
        "im_end":        text.count("<im_end>")
    }

def classify_case(raw_stats):
    if raw_stats["think_open"] == 1 and raw_stats["think_close"] == 1 \
        and raw_stats["answer_open"] == 1 and raw_stats["answer_close"] == 1 \
        and raw_stats["thinking_leak"] == 0:
        return "PASS_exact"
    elif raw_stats["thinking_leak"] > 0:
        return "FAIL_thinking_leak"
    elif raw_stats["think_open"] == 1 and raw_stats["answer_open"] == 0:
        return "PARTIAL_only_think"
    else:
        return "FAIL_no_schema"
```

7. Model Export, Quantization & GGUF Conversion

7.1 GGUF Container Format & Quantization Theory

GGUF (GGML Unified Format) is a binary file format designed for fast, single-file model loading on edge devices (CPU/GPU via `llama.cpp`). Quantization reduces 16-bit float weight representations down to lower bit-widths (4-bit, 5-bit, 8-bit) using block-wise scaling factor quantization:

$$w \approx s \cdot q + m$$

where q is the quantized k -bit integer, s is the block scale factor, and m is the zero-point offset.

7.2 The 2 Export Pipelines in Our Codebase

Pipeline A: Unsloth Direct Export (`07_modal_lily_1_5b_distill.ipynb`)

```
# Exports GGUF files directly from Unsloth engine
model_m.save_pretrained_gguf("gguf_model", tok_m, quantization_method="q4_k_m")
```

Pipeline B: Manual `llama.cpp` CLI Pipeline (`10_colab_lily_1_5b_v03_distill_gguf.ipynb`)

- 1. **System Prompt Patching:** Replaces base Qwen system templates with our standardized CoT system prompt across `config.json` and tokenizer files.
- 2. **F16 GGUF Conversion:** `bash python llama.cpp/convert_hf_to_gguf.py \ /content/Lily-1.5b-v0.3 \ --outtype f16 \ --outfile /content/Lily-1.5b-v0.3-F16.gguf`
- 3. **K-Quantization Execution:** `bash llama.cpp/build/bin/llama-quantize /content/Lily-1.5b-v0.3-F16.gguf /content/Lily-1.5b-v0.3-Q4_K_M.gguf Q4_K_M llama.cpp/build/bin/llama-quantize /content/Lily-1.5b-v0.3-F16.gguf /content/Lily-1.5b-v0.3-Q5_K_M.gguf Q5_K_M llama.cpp/build/bin/llama-quantize /content/Lily-1.5b-v0.3-F16.gguf /content/Lily-1.5b-v0.3-Q8_0.gguf Q8_0`

Quantization Variant Comparison Matrix

Variant	Quantization Method	File Size (1.5B Model)	VRAM Required	Recommended Usage
F16	Unquantized FP16	3.09 GB	~4.2 GB	Baseline reference evaluation
Q8_0	8-bit Standard Quant	1.62 GB	~2.5 GB	Server-side high-precision inference
Q5_K_M	5-bit K-Quant (Medium)	1.12 GB	~1.8 GB	Optimal quality/memory balance
Q4_K_M	4-bit K-Quant (Medium)	0.98 GB	~1.5 GB	Mobile / Edge CPU deployment

8. Multimodal Vision Extension (Lily Vision)

8.1 VLM Architecture & Dimensionality Pipeline

Our Vision-Language Model **Lily Vision** (`11` , `12` , `13`) connects a pre-trained Vision Tower to the Lily 1.5B LLM via a 2-layer projection bottleneck:

```
# Projector Architecture Implementation (12_lily-vision-phase1-alignment-v5.ipynb)
class LilyVisionProjector(nn.Module):
    def __init__(self, vision_dim=1152, llm_dim=1536):
        super().__init__()
        self.downsampler = nn.AdaptiveAvgPool2d((18, 18)) # 18x18 = 324 tokens
        self.mlp = nn.Sequential(
            nn.Linear(vision_dim, 2048),
            nn.SiLU(),
            nn.Linear(2048, llm_dim)
        )

    def forward(self, x): # x: [B, 729, 1152]
        B, N, C = x.shape
        H = W = int(N ** 0.5) # 27x27 grid
        x_2d = x.transpose(1, 2).view(B, C, H, W)
        x_pooled = self.downsampler(x_2d).flatten(2).transpose(1, 2) # [B, 324, 1152]
        return self.mlp(x_pooled) # [B, 324, 1536]
```

8.2 Phase 1 Alignment vs Phase 2 SFT — Layer Freezing Matrix

Architecture Submodule	Phase 1: Projector Alignment (12)	Phase 2: Multimodal SFT (13)
SigLIP-2 Vision Encoder	❄️ Frozen (requires_grad = False)	❄️ Frozen (requires_grad = False , set to .eval())
2-Layer MLP Projector	🔥 Trained (Random init, LR = 2×10^{-4})	🔥 Trained (From Phase 1 weights, LR = 2×10^{-5})
LLM Base Weights	❄️ Frozen (requires_grad = False)	❄️ Frozen (Loaded in 4-bit QLoRA)
LLM LoRA Adapters	N/A (No LoRA used)	🔥 Trained ($r = 32, \alpha = 64$ on 7 layers)
Visual Token Count	324 tokens (Pooled via 18x18 adaptive pool)	729 tokens (Full 27x27 patch resolution)
Sequence Length	256 tokens	3,072 tokens

8.3 LengthGroupedBatchSampler Performance Optimization

In 13_lily-vision-phase2-sft.ipynb , multimodal sequences vary drastically in token length. Standard batching causes massive padding overhead. We implemented a custom PyTorch batch sampler that groups sequences by length:

```
class LengthGroupedBatchSampler(Sampler):
    def __init__(self, lengths, batch_size, mega_batch_mult=50, seed=42):
        self.lengths = lengths
        self.batch_size = batch_size
        self.mega_batch_size = batch_size * mega_batch_mult # 24 * 50 = 1200 samples

    def __iter__(self):
        # Partition into mega-batches, sort by sequence length, form mini-batches
        pass
```

Optimization Outcome: Reduced total Phase 2 SFT training duration from ~3 hours down to ~20 minutes on an A100 40GB GPU.

9. Cloud Checkpointing & Fault-Tolerant Auto-Resumption

Across all cloud training notebooks (02 , 04 , 07 , 12 , 13), running on serverless infrastructure (Modal) presents a risk of preemption or timeout. We built custom Hugging Face Hub callbacks that automatically snapshot checkpoints to

private repos and auto-resume on restart.

```
from huggingface_hub import HfApi, snapshot_download

# Auto-Resume Logic (07_modal_lily_1_5b_distill.ipynb)
if RESUME_FROM_CHECKPOINT and CHECKPOINT_REPO:
    api = HfApi()
    files = list(api.list_repo_files(CHECKPOINT_REPO, token=HF_TOKEN))
    ckpt_nums = [int(f.split("/")[0].split("-")[-1]) for f in files if "checkpoint-" in f]
    if ckpt_nums:
        latest = max(ckpt_nums)
        local_dir = os.path.join(OUTPUT_DIR, f"checkpoint-{latest}")
        print(f"Downloading latest checkpoint-{latest} from HF Hub...")
        snapshot_download(
            repo_id = CHECKPOINT_REPO,
            allow_patterns = [f"checkpoint-{latest}/*"],
            local_dir = OUTPUT_DIR,
            token = HF_TOKEN,
        )
        _resume_ckpt = local_dir
```

Conclusion & Summary Checklist

By combining **PEFT (LoRA/QLoRA)**, **Hyperparameter Scaling Equations**, **GRPO Reinforcement Learning**, **Offline SFT Distillation**, **GGUF K-Quantization**, and **2-Phase Multimodal VLM Alignment**, our codebase establishes a complete, production-grade LLM development framework.